

Problem

You are given an array of N integers representing the weights of people.

Your task is to divide the people into two groups such that:

- The two groups have the same number of people, or their sizes differ by at most one. The absolute difference between the total weights of the two groups is minimized.

Print the minimum possible difference between the total weights of the two groups.

Submissions

Input Format

The first line contains an integer N, the number of people. The second line contains N space-separated integers representing their weights.

Leaderboard

Constraints

 $2 \leq N \leq 30$ $1 \leq \text{arr}[i] \leq 1000$

Output Format

Print a single integer representing the minimum possible difference between the total weights of the two groups.

Discussions

Sample Input 0

```
4
3 4 5 6
```

Sample Output 0

```
0
```

Sample Input 1

```
5
1 2 3 4 5
```

Sample Output 1

```
1
```

```
5 k=n//2
6 ans=10**9
7 def solve(i,c,s):
8     global ans
9     if c==k:
10         ans=min(ans,abs(total-2*s))
11         return
12     if i==n:
13         return
14     solve(i+1,c+1,s+a[i])
15     solve(i+1,c,s)
16 solve(0,0,0)
17 print(ans)
18
```

Line: 18 Col: 1



Upload Code as File

Run Code

Submit Code



Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?



Test case 0

Compiler Message



Test case 1

Success